

OTAR, Kazakhstan

WMO: 368640

Lat: 43.533N

Lon: 75.250E

Elev: 2438

StdP: 13.45

Time Zone: 6.00 (E06)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.5	-3.5	-15.4	2.6	-10.1	-8.9	3.7	-2.8	23.3	27.7	19.1	28.2	0.5	90	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.1	97.4	63.4	94.4	62.5	91.6	61.8	66.0	86.6	64.8	85.9	63.8	84.8	8.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.9	87.3	70.4	58.6	80.4	70.0	56.7	75.0	69.5	32.1	86.1	31.1	86.3	30.3	84.4	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	20.6	17.3	14.8	DB	-15.8	102.6	8.0	2.3	-21.5	104.2	-26.2	105.6	-30.7	106.9	-36.5	108.6
(5)				WB	-16.1	69.5	7.9	3.5	-21.8	72.0	-26.5	74.0	-30.9	76.0	-36.7	78.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	19.3	24.3	40.9	52.3	62.0	71.9	76.1	73.5	63.4	50.0	36.1	22.8
	DBStd	21.46	11.41	11.22	9.37	8.09	6.69	5.54	5.31	5.70	7.64	8.32	10.06	9.86
	HDD50	3415	952	720	300	66	3	0	0	0	3	103	424	844
	HDD65	6605	1417	1140	747	384	136	10	1	5	119	470	867	1309
	CDD50	3236	0	0	18	135	375	656	810	728	405	102	7	0
	CDD65	951	0	0	0	3	44	216	346	268	71	3	0	0
	CDH74	13649	0	0	1	100	762	2829	4786	3849	1231	91	0	0
	CDH80	6436	0	0	0	17	228	1251	2521	1921	481	17	0	0
Wind	WSAvg	4.8	2.9	3.6	5.4	6.0	6.4	6.0	5.8	5.7	5.1	4.3	3.6	3.2
Precipitation	PrecAvg	12.1	0.7	0.8	1.0	1.5	1.5	1.1	1.1	0.5	0.6	1.2	1.2	0.9
	PrecMax	19.4	2.3	2.4	1.9	3.3	3.6	2.7	2.7	1.6	1.9	3.5	3.0	1.7
	PrecMin	7.6	0.2	0.3	0.4	0.2	0.2	0.2	0.2	0.0	0.0	0.0	0.2	0.3
	PrecStd	3.0	0.4	0.4	0.4	0.8	0.8	0.7	0.6	0.5	0.5	0.9	0.8	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	50.2	70.9	82.5	90.5	97.6	102.6	99.9	93.9	82.4	65.3	46.5
		MCWB	38.6	42.3	52.7	56.7	60.4	62.9	64.5	64.1	60.0	55.8	51.7	39.6
	2%	DB	39.0	44.2	65.3	76.6	86.1	93.7	98.4	96.7	89.5	75.6	58.8	40.8
		MCWB	34.9	38.4	50.4	56.5	59.7	62.4	64.2	62.7	58.8	53.8	48.4	36.3
	5%	DB	36.2	40.5	60.2	72.4	82.0	90.7	95.4	93.7	85.8	70.6	54.0	37.9
		MCWB	33.1	36.1	48.6	55.1	59.0	61.7	63.4	62.0	57.6	51.7	45.5	34.2
	10%	DB	33.7	37.5	55.8	68.0	78.2	87.3	92.2	90.1	81.5	65.5	49.9	35.3
		MCWB	31.3	34.0	46.6	53.3	57.7	61.2	62.7	61.1	56.3	49.6	44.0	32.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.9	43.2	54.9	61.7	65.1	66.8	68.2	66.6	62.4	57.5	53.3	40.6
		MCDB	44.3	49.1	65.9	75.1	80.9	85.0	86.9	87.9	83.7	76.2	62.7	44.5
	2%	WB	35.7	39.1	51.7	58.6	62.7	65.3	66.4	64.7	60.4	55.4	49.8	36.8
		MCDB	38.5	43.7	62.8	71.3	78.6	83.2	87.8	88.5	83.9	70.6	56.6	40.1
	5%	WB	33.5	36.3	49.2	56.8	60.8	64.1	65.2	63.6	58.8	53.5	46.7	34.6
		MCDB	35.7	40.0	58.6	69.5	76.3	82.0	87.4	87.3	81.3	66.7	52.6	37.4
	10%	WB	31.5	34.2	47.1	54.7	59.3	63.0	64.0	62.4	57.3	51.5	43.8	32.5
		MCDB	33.6	37.1	55.4	65.8	74.5	81.2	86.4	85.5	78.9	62.7	49.5	35.0
Mean Daily Temperature Range	5% DB	MDBR	16.3	15.4	18.9	21.6	25.1	26.1	27.1	28.5	28.4	23.1	16.7	15.3
		MCDBR	18.0	18.1	27.1	30.5	31.5	31.8	33.2	33.8	34.7	32.2	25.2	18.9
		MCWBR	14.1	13.0	14.8	14.1	11.7	9.7	10.5	11.5	13.5	14.5	15.2	14.6
	5% WB	MCDBR	15.4	16.5	24.2	26.6	26.1	26.5	27.3	29.4	30.0	25.7	22.2	16.4
		MCWBR	12.4	12.2	14.0	13.2	11.3	8.8	8.9	10.2	12.2	12.2	13.9	12.9
Clear-Sky Solar Irradiance	taub	0.363	0.413	0.473	0.502	0.475	0.472	0.464	0.429	0.411	0.439	0.410	0.357	
	taud	2.039	1.953	1.873	1.850	1.976	2.029	2.067	2.147	2.132	1.970	1.983	2.048	
	Ebn at Noon	227	235	238	243	255	256	256	262	256	223	205	214	
	Edh at Noon	35	45	56	62	56	53	51	45	43	44	37	31	
All-Sky Solar Radiation	RadAvg	627	911	1209	1615	2071	2193	2183	2024	1609	1025	612	482	
	RadStd	68	100	101	147	142	105	132	100	71	95	59	44	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	-0.86	-1.64	N/A	N/A	N/A	N/A

Nomenclature: See separate page